Experiment 1

Aim:

Write a program in C to read 10 numbers from keyboard and find their sum and average.

Program:

```
#include <stdio.h>
void main()
{
    int i,n,sum=0;
■float avg;
■printf("Input the 10 numbers : \n");
■for (i=1;i<=10;i++)
■{
                printf("Number-%d :",i);

■■scanf("%d",&n);
■■sum +=n;
■}
■avg=sum/10.0;
■printf("The sum of 10 no is : %d\nThe Average is : %f\n",sum,avg);

}
```

Output:

Run the program and display the result as shown in the console.

Experiment 2

Aim:

Write a program in C to display the cube of the number upto a given integer.

Program:

```
#include <stdio.h>
void main()
 {
    int i,ctr;
    printf("Input number of terms : ");
    scanf("%d", &ctr);
    for(i=1;i<=ctr;i++)
    {
■ printf("Number is : %d and cube of the %d is :%d \n",i,i, (i*i*i));
    }
 }
```

Output:

Run the program and display the result as shown in the console.

Experiment 3

Aim:

Write a program in C to display the multiplication table of a given integer

Program:

```
#include <stdio.h>
void main()
{
   int j,n;
   printf("Input the number (Table to be calculated) : ");
   scanf("%d",&n);
   printf("\n");
   for(j=1;j<=10;j++)
   {
     printf("%d X %d = %d \n",n,j,n*j);
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 4

Aim:

Write a C program to calculate the factorial of a given number.

Program:

```
#include <stdio.h>
void main(){
  int i,f=1,num;

  printf("Input the number : ");
  scanf("%d",&num);

  for(i=1;i<=num;i++)
      f=f*i;

  printf("The Factorial of %d is: %d\n",num,f);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 5

Aim:

Write a program in C to display the n terms of harmonic series and their sum. The series is :

Program:

```
#include <stdio.h>
void main()
{
   int i,n;
   float s=0.0;
   printf("Input the number of terms : ");
   scanf("%d",&n);
   printf("\n\n");
   for(i=1;i<=n;i++)
   {
       if(i<n)
       {
     printf("1/%d + ",i);
     s+=1/(float)i;
       }
     if(i==n)
     {
     printf("1/%d ",i);
     s+=1/(float)i;
     }
     }
        printf("\nSum of Series upto %d terms : %f \n",n,s);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 6

Aim:

Write a program in C to find the sum of the series 1 +11 + 111 + 1111 + .. n terms.

Program:

```
#include <stdio.h>

void main()
{
  int n,i;
  long sum=0;
  long int t=1;
  printf("Input the number of terms : ");
  scanf("%d",&n);
  for(i=1;i<=n;i++)
  {
     printf("%ld  ",t);
      if (i<n)
      {
          printf("+ ");

      }
     sum=sum+t;
     t=(t*10)+1;
  }
  printf("\nThe Sum is : %ld\n",sum);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 7

Aim:

Write a C program to determine whether a given number is prime or not.

Program:

```
#include <stdio.h>

void main(){

    int num,i,ctr=0;
    printf("Input  a number: ");
    scanf("%d",&num);
    for(i=2;i<=num/2;i++){
        if(num % i==0){
         ctr++;
            break;
        }
    }
   if(ctr==0 && num!= 1)
        printf("%d is a prime number.\n",num);
   else
      printf("%d is not a prime number",num);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 8

Aim:

Write a program in C to display the first n terms of Fibonacci series. The series is as follows:

Program:

```
#include <stdio.h>

void main()
{
   int prv=0,pre=1,trm,i,n;
   printf("Input number of terms to  display : ");
   scanf("%d",&n);
   printf("Here is the Fibonacci series upto  to %d terms : \n",n);
   printf("% 5d % 5d", prv,pre);

  for(i=3;i<=n;i++)
   {
     trm=prv+pre;
     printf("% 5d",trm);
     prv=pre;
     pre=trm;
   }
   printf("\n");
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 9

Aim:

Write a program in C to display the multiplication table vertically from 1 to n.

Program:

```
#include <stdio.h>
void main()
{
   int j,i,n;
   printf("Input upto the table number starting from 1 : ");
   scanf("%d",&n);
   printf("Multiplication table from 1 to %d \n",n);
   for(i=1;i<=10;i++)
   {
     for(j=1;j<=n;j++)
     {
       if (j<=n-1)
           printf("%dx%d = %d, ",j,i,i*j);
          else
■    printf("%dx%d = %d",j,i,i*j);

      }
     printf("\n");
    }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 10

Aim:

Write a program in C to display the pattern like right angle triangle using an asterisk.

Program:

```
#include <stdio.h>
void main()
{
   int i,j,rows;
   printf("Input number of rows : ");
   scanf("%d",&rows);
   for(i=1;i<=rows;i++)
   {
■for(j=1;j<=i;j++)
■   printf("*");
■printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 11

Aim:

Write a program in C to display the pattern like right angle triangle with a number.

Program:

```
#include <stdio.h>
void main()
{
   int i,j,rows;
   printf("Input number of rows : ");
   scanf("%d",&rows);
   for(i=1;i<=rows;i++)
   {
■for(j=1;j<=i;j++)
■   printf("%d",j);
■printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 12

Aim:

Write a program in C to make such a pattern like right angle triangle with a number which will repeat a number in a row.

Program:

```
#include <stdio.h>
void main()
{
   int i,j,rows;

   printf("Input number of rows : ");
   scanf("%d",&rows);
   for(i=1;i<=rows;i++)
   {
■for(j=1;j<=i;j++)
■   printf("%d",i);
■printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 13

Aim:

Write a program in C to make such a pattern like right angle triangle with number increased by 1.

Program:

```
#include <stdio.h>
void main()
{
   int i,j,rows,k=1;
   printf("Input number of rows : ");
   scanf("%d",&rows);
   for(i=1;i<=rows;i++)
   {
■for(j=1;j<=i;j++)
■   printf("%d ",k++);
■printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 14

Aim:

Write a program in C to make such a pattern like a pyramid with numbers increased by 1. The pattern is as follows:

Program:

```
#include <stdio.h>
void main()
{
   int i,j,spc,rows,k,t=1;
   printf("Input number of rows : ");
   scanf("%d",&rows);
   spc=rows+4-1;
   for(i=1;i<=rows;i++)
   {
         for(k=spc;k>=1;k--)
            {
              printf(" ");
            }
■    for(j=1;j<=i;j++)
■    printf("%d ",t++);
■printf("\n");
    spc--;
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 15

Aim:

Write a program in C to make such a pattern like a pyramid with an asterisk. The pattern is as follows :

Program:

```
#include <stdio.h>
void main()
{
   int i,j,spc,rows,k;
   printf("Input number of rows : ");
   scanf("%d",&rows);
   spc=rows+4-1;
   for(i=1;i<=rows;i++)
   {
         for(k=spc;k>=1;k--)
            {
              printf(" ");
            }

■    for(j=1;j<=i;j++)
■    printf("* ");
■printf("\n");
    spc--;
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 16

Aim:

Write a program in C to make such a pattern like a pyramid with a number which will repeat the number in the same row.

Program:

```
#include <stdio.h>

void main()
{
   int i,j,spc,rows,k;
   printf("Input number of rows : ");
   scanf("%d",&rows);
   spc=rows+4-1;
   for(i=1;i<=rows;i++)
   {
         for(k=spc;k>=1;k--)
            {
              printf(" ");
            }

■   for(j=1;j<=i;j++)
■   printf("%d ",i);
■printf("\n");
    spc--;
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 17

Aim:

Write a program in C to print the Floyd's Triangle. The Floyd's triangle is as below :

Program:

```
#include <stdio.h>

void main()
{
   int i,j,n,p,q;
   printf("Input number of rows : ");
   scanf("%d",&n);
   for(i=1;i<=n;i++)
   {
     if(i%2==0)
     { p=1;q=0;}
     else
     { p=0;q=1;}
      for(j=1;j<=i;j++)
■ if(j%2==0)
■    printf("%d",p);
■ else
■    printf("%d",q);
     printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 18

Aim:

Write a program in C to display the pattern like a diamond. The pattern is as follows :

Program:

```
#include <stdio.h>

void main()
{
   int i,j,r;
   printf("Input number of rows (half of the diamond) :");
   scanf("%d",&r);
   for(i=0;i<=r;i++)
   {
     for(j=1;j<=r-i;j++)
     printf(" ");
     for(j=1;j<=2*i-1;j++)
       printf("*");
     printf("\n");
   }

   for(i=r-1;i>=1;i--)
   {
     for(j=1;j<=r-i;j++)
     printf(" ");
     for(j=1;j<=2*i-1;j++)
       printf("*");
     printf("\n");
   }

}
```

Output:

Run the program and display the result as shown in the console.

Experiment 19

Aim:

Write a C program to display Pascal's triangle

Program:

```
#include <stdio.h>

void main()
{
    int no_row,c=1,blk,i,j;
    printf("Input number of rows: ");
    scanf("%d",&no_row);
    for(i=0;i<no_row;i++)
    {
        for(blk=1;blk<=no_row-i;blk++)
        printf("  ");
        for(j=0;j<=i;j++)
        {
            if (j==0||i==0)
                c=1;
            else
               c=c*(i-j+1)/j;
            printf("% 4d",c);
        }
        printf("\n");
    }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 20

Aim:

Write a program in C to display the such a pattern for n number of rows using a number which will start with the number 1 and the first and a last number of each row will be 1. The pattern is as follows:

Program:

```
#include <stdio.h>

void main()
{
   int i,j,n;
   printf("Input number of rows : ");
   scanf("%d",&n);
   for(i=0;i<=n;i++)
   {
     /* print blank spaces */
     for(j=1;j<=n-i;j++)
■printf(" ");
     /* Display number in ascending order upto middle*/
     for(j=1;j<=i;j++)
       printf("%d",j);

     /* Display  number in reverse order after middle */
       for(j=i-1;j>=1;j--)
■   printf("%d",j);

     printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 21

Aim:

Write a C Program to display the pattern like pyramid using the alphabet. The pattern is as follows :

Program:

```
#include <stdio.h>

void main()
  {
   int i, j;
   char alph = 'A';
   int n,blk;
   int ctr = 1;

   printf("Input the number of Letters (less than 26) in the Pyramid : ");
   scanf("%d", &n);

   for (i = 1; i <= n; i++)
■{
■for(blk=1;blk<=n-i;blk++)

■printf("  ");
      for (j = 0; j <= (ctr / 2); j++) {
         printf("%c ", alph++);
      }

      alph = alph - 2;

      for (j = 0; j < (ctr / 2); j++) {
         printf("%c ", alph--);
      }
      ctr = ctr + 2;
      alph = 'A';
      printf("\n");
   }
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 22

Aim:

Write a program in C to find the sum of the series [1-X^2/2!+X^4/4!- ....].

Program:

```
#include <stdio.h>
void main()
{
■float x,sum,t,d;
■int i,n;
■printf("Input the Value of x :");
■scanf("%f",&x);
■printf("Input the number of terms : ");
■scanf("%d",&n);
■sum =1; t = 1;
■for (i=1;i<n;i++)
■{
■  d = (2*i)*(2*i-1);
■  t = -t*x*x/d;
■  sum =sum+ t;
■}
■printf("\nthe sum = %f\nNumber of terms = %d\nvalue of x = %f\n",sum,n,x);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 23

Aim:

Write a program in C to display the sum of the series [1+x+x^2/2!+x^3/3!+....].

Program:

```
#include <stdio.h>

void main()
{
■float x,sum,no_row;
■int i,n;
■printf("Input the value of x :");
■scanf("%f",&x);
■printf("Input number of terms : ");
■scanf("%d",&n);
■sum =1; no_row = 1;
■for (i=1;i<n;i++)
■{
■  no_row = no_row*x/(float)i;
■  sum =sum+ no_row;
■}
■printf("\nThe sum  is : %f\n",sum);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 24

Aim:

Write a program in C to find the sum of the series [x - x^3 + x^5 + ...].

Program:

```
#include <stdio.h>
#include <math.h>
void main()
{
■int x,sum,ctr;
■int i,n,m,mm,nn;
■printf("Input the value of x :");
■scanf("%d",&x);
■printf("Input number of terms : ");
■scanf("%d",&n);
■sum =x; m=-1;
■printf("The values of the series: \n");
■printf("%d\n",x);
    for (i = 1; i < n; i++)
    {
        ctr = (2 * i + 1);
        mm = pow(x, ctr);
        nn = mm * m;
        printf("%d \n",nn);
        sum = sum + nn;
        m = m * (-1);
■}
■printf("\nThe sum = %d\n",sum);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 25

Aim:

Write a program in C to find the number and sum of all integer between 100 and 200 which are divisible by 9.

Program:

```
#include <stdio.h>

void main()
{
   int i, sum=0;
   printf("Numbers between 100 and 200, divisible by 9 : \n");
   for(i=101;i<200;i++)
   {
     if(i%9==0)
     {
       printf("% 5d",i);
       sum+=i;
     }
   }
   printf("\n\nThe sum : %d \n",sum);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 26

Aim:

Write a C program to find HCF (Highest Common Factor) of two numbers.

Program:

```
#include <stdio.h>

void main()
{
    int i, n1, n2, j, hcf=1;


     printf("\n\n  HCF of two numbers:\n ");
     printf("----------------------\n");


    printf("Input 1st number for HCF: ");
    scanf("%d", &n1);
    printf("Input 2nd number for HCF: ");
    scanf("%d", &n2);

    j = (n1<n2) ? n1 : n2;

    for(i=1; i<=j; i++)
    {

        if(n1%i==0 && n2%i==0)
        {
            hcf = i;
        }
    }

    printf("\nHCF of %d and %d is : %d\n\n", n1, n2, hcf);
}
```

Output:

Run the program and display the result as shown in the console.

Experiment 27

Aim:

Write a program in C to find LCM of any two numbers using HCF

Program:

```
#include <stdio.h>

void main()
{
    int i, n1, n2, j, hcf=1,lcm;


     printf("\n\n  LCM of two numbers:\n ");
     printf("----------------------\n");


    printf("Input 1st number for LCM: ");
    scanf("%d", &n1);
    printf("Input 2nd number for LCM: ");
    scanf("%d", &n2);

    j = (n1<n2) ? n1 : n2;

    for(i=1; i<=j; i++)
    {

        if(n1%i==0 && n2%i==0)
        {
            hcf = i;
        }
    }
/* We know  the multiplication of HCF and LCM is equivalant
to the multiplication of these two numbers.*/
    lcm=(n1*n2)/hcf;

    printf("\nThe LCM of %d and %d is : %d\n\n", n1, n2, lcm);

}
```

Output:

Run the program and display the result as shown in the console.

Experiment 28

Aim:

Write a program in C to find LCM of any two numbers.

Program:

```
#include <stdio.h>

void main()
{
    int i, n1, n2, max, lcm=1;


     printf("\n\n  LCM of two numbers:\n ");
     printf("----------------------\n");


    printf("Input 1st number for LCM: ");
    scanf("%d", &n1);
    printf("Input 2nd number for LCM: ");
    scanf("%d", &n2);


    max = (n1>n2) ? n1 : n2;


    for(i=max;  ; i+=max)
    {

        if(i%n1==0 && i%n2==0)
        {
            lcm = i;
            break;
        }
    }

    printf("\nLCM of %d and %d = %d\n\n", n1, n2, lcm);

}
```

Output:

Run the program and display the result as shown in the console.

Experiment 29

Aim:

Write a program in C to convert a binary number into a decimal number using math function.

Program:

```
#include <stdio.h>
#include <math.h>
void main()
{       int n1, n;
■int dec=0,i=0,j,d;

     printf("\n\nConvert Binary to Decimal:\n ");
     printf("-------------------------\n");


■printf("Input  the binary number :");
■scanf("%d",&n);
■n1=n;
■while(n!=0)
■{  d = n % 10;
■   dec=dec+d*pow(2,i);
■   n=n/10;
■   i++;
■}
        printf("\nThe Binary Number : %d\nThe equivalent Decimal  Number is : %d\n\n",n1,dec);
}
```

Output:

Run the program and display the result as shown in the console.